Mathematics - Formula sheet

⇒ ***Law of indices***:

- $a^m \, x \, a^n = a^{m+n}$
- $a^m \div a^n = a^{m-n}$
- $(a^m)^n = a^{mn}$
- $(ab)^n = a^n b^n$
- $a^{\frac{1}{m}} = \sqrt[m]{a}$
- $a^{\frac{n}{m}} = \sqrt[m]{a^n}$
- $a^{-m} = \frac{1}{a^m}$
- $a^0 = 1$

⇒ $x^2 - y^2 = (x+y)(x-y)$

⇒ $\sqrt{ab} = \sqrt{a} \, x \, \sqrt{b}$

⇒ $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$

⇒ $\frac{1}{\sqrt{a}}$; *multiply numerator and denom − inator by* $\sqrt{a}$

⇒ $\frac{1}{a+\sqrt{b}}$; *multiply numerator and denom − inator by* $(a - \sqrt{b})$

⇒ $\frac{1}{a-\sqrt{b}}$; *multiply numerator and denom − inator by* $(a + \sqrt{b})$

⇒ ***Completing the square***:

- $x^2 + bx = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2$
- $ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 - \left(\frac{b^2}{4a}\right) + c$

$\boldsymbol{f(x) = ax^2 + bx + c}$

- *when a is postive, the parabola shape is*: ∪
- *when a is negative, the parabola shape is*: ∩

$\boldsymbol{f(x) = a(x+p)^2 + q}$

- *turning point* $= (-p, q)$

$$\boldsymbol{\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}}$$

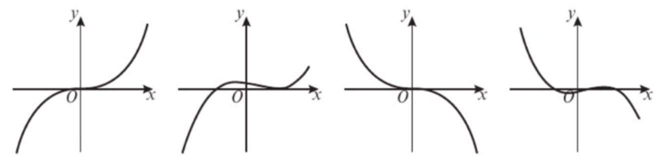
- $b^2 - 4ac > 0$; *two distinct real roots*
- $b^2 - 4ac = 0$; *one repeated root*
- $b^2 - 4ac < 0$; *no real roots*

⇒ *used for* $<>$, *end value is not included (Dotted line).*

⇒ *used for* $\leq \geq$, *end value is included (Solid line).*

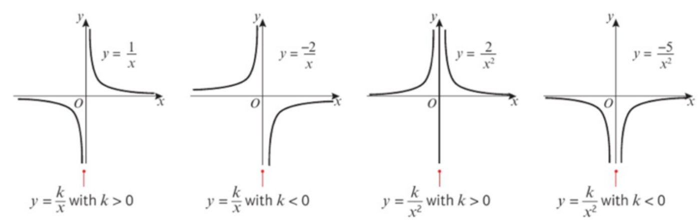
⇒ *Set notation* $= \{x : x \; [sign(<, >, \leq, \geq)] \; (value)\}$

⇒ ***Cubic function*** −
$f(x) = ax^3 + bx^2 + cx + d$



For these two functions a is positive. For these two functions a is negative.

⇒ ***Reciprocal graph***



⇒ *To find points of intersection of two graphs, equate both equations and find the solutions.*

Translation:

- *For* $y = f(x) + a$, *graph moves a units vertically (y − axis) up.*
- *For* $y = f(x + a)$, *graph moves a units horizontally (x − axis) left.*

Stretching:

- *For* $y = af(x)$, *graph stretches by a scale factor of a vertically (y − axis).*
- *For* $y = f(ax)$, *graph stretches by a scale factor of* $\frac{1}{a}$ *horizontally (x − axis).*

Reflection:

- *For* $y = f(-x)$, *graph reflects from y − axis.*
- *For* $y = -f(x)$, *graph reflects from x − axis.*

⇒ *Gradient(m)* $= \frac{y_2 - y_1}{x_2 - x_1}$

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- $\Rightarrow y = mx + c$
- $\Rightarrow ax + by + c = 0$
- $\Rightarrow y - y_1 = m(x - x_1)$
- $\Rightarrow$ *Parallel lines have same gradient.*
- $\Rightarrow$ *If a line has gradient m, the line perpendicular to it has gradient* $-\frac{1}{m}$
- $\Rightarrow$ *If two lines are perpendicular, their gradient is* $m_1 x\, m_2 = -1$.
- $\Rightarrow$ ***Distance between two points*** $= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- $\Rightarrow$ ***Midpoint*** $= \left[\frac{x1+x2}{2}, \frac{y1+y2}{2}\right]$

Cosine rule −

- $a^2 = b^2 + c^2 - 2bc\cos A$
- $Cos\ A = \frac{b^2+c^2-a^2}{2bc}$

Sine rule −

- $\frac{Sin\ A}{a} = \frac{Sin\ B}{b} = \frac{Sin\ C}{c}$
- $\Rightarrow \sin\theta = \frac{Opposite}{Hypotenuse}$
- $\Rightarrow \tan\theta = \frac{Opposite}{Adjacent}$
- $\Rightarrow \cos\theta = \frac{Adjacent}{Hypotenuse}$

- $f(x) = x^n\, ; f'(x) = nx^{n-1}$
- $f(x) = ax^n\, ; f'(x) = anx^{n-1}$
- ⇨ ***For quadratic curve*** $\boldsymbol{y = ax^2 + bx + c}$: $\frac{dy}{dx} = 2ax + b$
- ⇨ ***For curve*** $\boldsymbol{y = f(x)}$, ***the gradient of the tangent at point A with x − coordinate a will be*** $\boldsymbol{f'(a)}$; $(\boldsymbol{a}, \boldsymbol{f(a)})$
- ⇨ $y - f(a) = f'(a)(x - a)$
- ⇨ ***The normal to a curve at point A is the straight line through a perpendicular to the tangent to curve at A.***
- ⇨ $y - f(a) = -\frac{1}{f'(a)}(x - a)$
- ⇨ *gradient of normal* $= \frac{1}{f'(a)}$

Two possible solutions for a missing angle (sine rule) −

- $\sin\theta = \sin(180 - \theta)$
- $\Rightarrow$ *Area of triangle* $= \frac{1}{2}ab\sin C$
  - $\Rightarrow 2\pi$ *radians* $= 360°$
  - $\Rightarrow \pi$ *radians* $= 180°$
  - $\Rightarrow 1$ *radians* $= \frac{180°}{\pi}$
  - $\Rightarrow$ ***To convert radains to degrees −*** *Multiply by* $\frac{180°}{\pi}$
  - $\Rightarrow$ ***To convert degrees to radians −*** *Multiply by* $\frac{\pi}{180°}$
  - $\Rightarrow$ ***Arc length of sector −*** $l = r\theta\ (rad)$
  - $\Rightarrow$ ***Area of sector −*** $A = \frac{1}{2}r^2\theta\ (rad)$
  - $\Rightarrow$ ***Area of segment −*** $A = \frac{1}{2}r^2(\theta - sin\theta(rad))$
- ⇨ ***Gradient of curve***:

$$f'(x) = \lim_{h\to 0}\frac{f(x+h) - f(x)}{h}$$

Integration is the reverse process of differentiation.

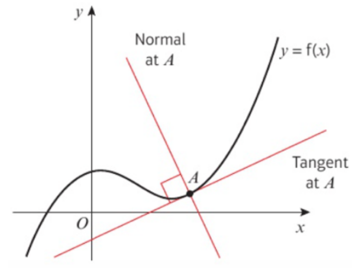
- $x^n = \frac{x^{n+1}}{n+1}$
- $\frac{dy}{dx} = \frac{x^{n+1}}{n+1} + c$
- $\frac{dy}{dx}$ *of* $kxn = \frac{k}{n+1}x^{n+1} + c$
- $\int x^n dx = \frac{x^{n+1}}{n+1} + c$

$\boldsymbol{To\ find\ constant\ of\ integration, c}$:

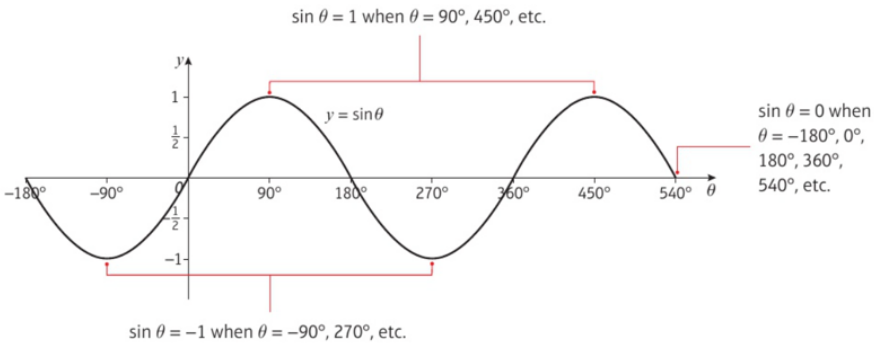
⇨ $Integrate\ the\ function.$

⇨ $Subsitute\ the\ values\ (x, y)\ of\ a\ point\ on\ the\ curve.$

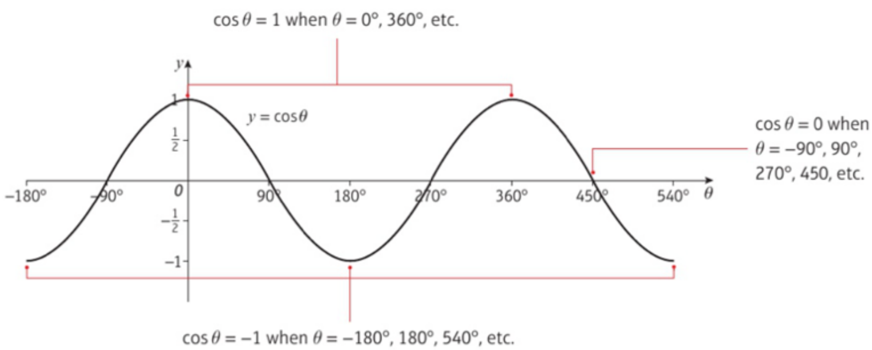
⇨ $Solve\ the\ equation\ to\ find\ c.$



- The graph of $y = \sin\theta$:
  - repeats itself every 360° period and crosses the x-axis at …, –180°, 0°, 180°, 360°, …
  - has a maximum value of 1 and a minimum value of –1.

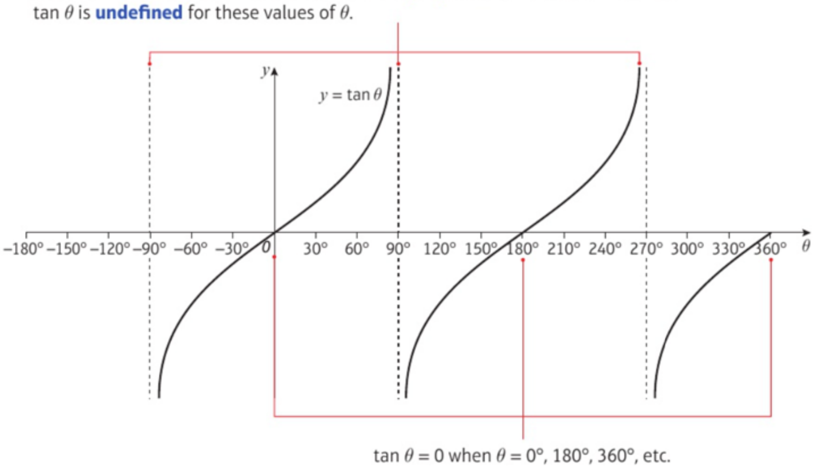


- The graph of $y = \cos\theta$:
  - repeats itself every 360° period and crosses the x-axis at …, –90°, 90°, 270°, 450°, …
  - has a maximum value of 1 and a minimum value of –1.



- The graph of $y = \tan\theta$:
  - repeats itself every 180° period and crosses the x-axis at … –180°, 0°, 180°, 360°, …
  - has no maximum or minimum value
  - has vertical asymptotes at $x = -90°$, $x = 90°$, $x = 270°$, …

tan θ does *not* have maximum and minimum points but approaches negative or positive infinity as the curve approaches the **asymptotes** at –90°, 90°, 270°, etc. tan θ is **undefined** for these values of θ.



Pure mathematics 2 (formula sheet)

⇒ ***Factor theorem***:

- $If\ f(p) = 0, then\ (x-p)\ is\ a\ factor\ of\ f(x)$
- $If\ (x-p) is\ a\ factor\ of\ f(x), then\ f(p) = 0$
- $If\ f\left(\frac{b}{a}\right) = 0, then\ (ax-b)\ is\ a\ factor\ of\ f(x).$
- $If\ (ax-b)\ is\ a\ factor\ of\ f(x), then\ f\left(\frac{b}{a}\right) = 0$

⇒ ***Remainder theorem***:

- $If\ f(x)\ is\ divided\ by\ (ax-b), then\ the$

$remainder\ is\ f\left(\frac{b}{a}\right)$

⇒ ***Methods of proof***

○ **Proof by exhaustion** – *breaking the statement into smaller cases and solving each case separately.*

- *For example*:

For odd numbers,

$(2n+1)^2 = 4n^2 + 4n + 1 = 4n(n+1) + 1$
$4n(n+1)\ is\ a\ multiple\ of\ 4\ so\ 4n+1\ is$
$one\ more\ than\ a\ multiple\ of\ 4.$

○ **Disproof by Counter** – *This method shows that the statement made is not true for every possible case and is, therefore, incorrect.*

- *For example*:

Show that the following statement is false:
"(n + 1)^3 − n^3 is prime for all natural numbers"
When n = 5, the answer is 91, which has factors of 7 and 13 and is therefore not prime. Thereby sufficiently disproving the statement.

⇒ ***Midpoint of straight line***: $\left[\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right]$

⇒ $m_1\ x\ m_x = -1$

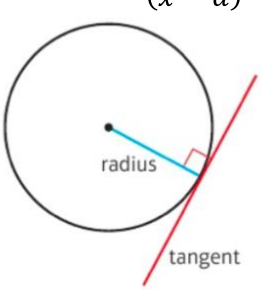
⇒ ***Equation of circle –***

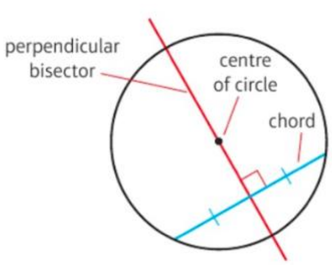
⇒ *With centre (0,0) and radius r is:*

$$x^2 + y^2 = r^2$$

⇒ *With centre (a, b) and radius r is:*

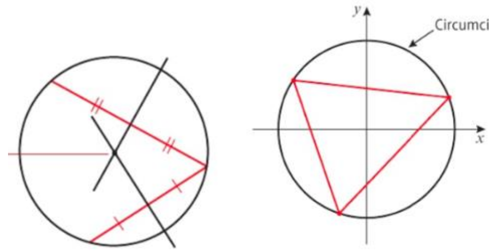
$$(x-a)^2 + (y-b)^2 = r^2$$





⇒ ***To find center of circle given any three points on the circumference***:

- *Find equations of the perpendicular bisectors of two different chords.*
- *Find the points of intersection by equating the equations.*



⇒ ***Exponential functions are in the form*** $\boldsymbol{f(x) = a^x}$.

- *When* $a > 1, f(x)$ *is an increasing function,*
- *When* $0 < a < 1, f(x)$ *is a decreasing function.*

⇒ $\boldsymbol{\log_a n = x}$ ***is equivalent to*** $\boldsymbol{a^x = n}$

⇒ ***Laws of logarithms***:

- $\log_a x + \log_a y = \log_a xy$ *(multiplication law)*
- $\log_a x - \log_a y = \log_a \left(\frac{x}{y}\right)$ *(division law)*
- $\log_a (x^k) = k \log_a x$ *(power law)*

⇒ ***Laws for special cases***:

- $\log_a \left(\frac{1}{x}\right) = \log_a (x^{-1}) = -\log_a x$ *(when k = −1)*
- $\log_a a = 1$
- $\log_a 1 = 0$
- $\log_a b = \frac{1}{\log_b a}$
- $\log_a x = \frac{\log_b x}{\log_b a}$

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
1 6 15 20 15 6 1
1 7 21 35 35 21 7 1

⇒ *The (n + 1)th row of* ***pascals triangle*** *gives the coefficients in the expansion* $(a+b)^n$

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⇒ ***Factorial notation*** –

- $nCr = \binom{n}{r} = \frac{n!}{r!(n-r)!}$
- ${}^{n-1}C_{r-1} = \binom{n-1}{r-1}$

⇒ ***Binomial expansion***:

- $(a+b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \cdots + \binom{n}{r}a^{n-r}b^r + b^n$ *where* $\binom{n}{r} = nCr = \frac{n!}{r!(n-r)!}$

⇒ ***Arithmetic sequence*** *have constant difference between consecutive numbers.*

- *Formula*: $a + (n-1)d$
  $a = \text{first term}$
  $d = \text{common difference}$

⇒ ***Arithmetic series*** *is the sum of the terms of an arithmetic sequence.*

- *Formula for sum of first n terms of an arithmetic series*:

○ $S_n = \frac{n}{2}\big(2a + d(n-1)\big)$
{$a = \text{first term}; d = \text{common difference}$)

○ $S_n = \frac{n}{2}(a+l)$ {$l = \text{last tine}$)

⇒ ***Geometric sequence*** *have common ration between consecutive terms.*

- *To get from one term to the next, you multiply by the common ratio.*
  $u_n = ar^{n-1}$ {$r = \text{common ratio}$}

⇒ ***Geometric series*** *is the sum of the terms of a geometric sequence.*

- *Formula for sum of first n terms of an geometric series*:
  ○ $S_n = \frac{a(1-r^n)}{1-r}$ {*Easier to use if* $r < 1$}
  ○ $S_n = \frac{a(r^n-1)}{r-1}$ {*Easier to use if* $r > 1$}

⇒ ***Sum to infinity*** – *n tends to infinity*

⇒ ***Divergent series*** – (*Terms of the sequence increase*) *as n tend to infinity, S_n also tends to infinity.*

⇒ ***Convergent series*** – (*Terms of the sequence decrease*) *as n tend to infinity,* S_n *also tends to infinity.*

⇒ *A geometric sequence is convergent only when* $-1 < r < 1$ {$|r|$}.

⇒ ***Sum of first n terms of a geometric sequence***:

- $|r| < 1, \lim_{n\to\infty}\left(\frac{a(1-r^n)}{1-r}\right) = \frac{a}{1-r}$

⇒ ***Sum to infinity of a convergent geometric series***:

- $S_\infty = \frac{a}{1-r}$

⇒ ***Sigma notation*** – *The limit is written on the top and the terms you are summing are written on the bottom.*

- $\sum_{r=1}^{n} 1 = n$
- $\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$
- $\sum_{r=k}^{n} u_r = \sum_{r=1}^{n} u_r - \sum_{r=1}^{k-1} u_r$

⇒ ***Reccurrence relation*** *of the form*:

$$u_{n+1} = f(u_n)$$

defines each term of a seqence as a function of the previous term.

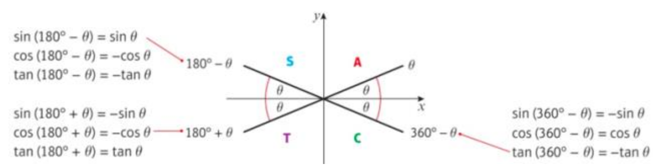
- *Sequence increases if* $u_{n+1} > u_n$
- *Sequence decreases if* $u_{n+1} < u_n$

⇒ *A Sequence is* ***periodic*** *if the terms repeat in a cycle.*

$u_{n+k} > u_n$ {*k is the order of the sequence*}

⇒ ***For a point $P(x,y)$ on a unit circle such that OP makes an angle with the positive − axis***:

- $\cos\theta = x - axis$ {$x - \text{coordinate of } OP$}
- $\sin\theta = y - axis$ {$y - \text{coordinate of } OP$}
- $\tan\theta = \frac{y-axis}{x-axis}$ {*gradient of OP*}



⇒ $\boldsymbol{sin^2\theta + cos^2\theta = 1}$

⇒ $\boldsymbol{\tan\theta = \frac{\sin\theta}{\cos\theta}}$

⇒ Solutions to $\sin\theta = k$ *and* $\cos\theta = k$ *only exist when* $-1 \le k \le 1$

⇒ Solutions to $\tan\theta = p$ *exist for all values of p*

⇒ ***Calculator will give principal values in the following ranges***:

- $\cos^{-1}$ *in the range* $0 \le \theta \le 180^o$
- $\sin^{-1}$ *in the range* $-90 \le \theta \le 90^o$
- $\tan^{-1}$ *in the range* $-90 \le \theta \le 90^o$

⇒ ***The function $f(x)$ is increasing on the interval $[a,b]$ if $f'(x) \ge 0$.***

⇒ ***The function $f(x)$ is decreasing on the interval $[a,b]$ if $f'(x) \le 0$.***

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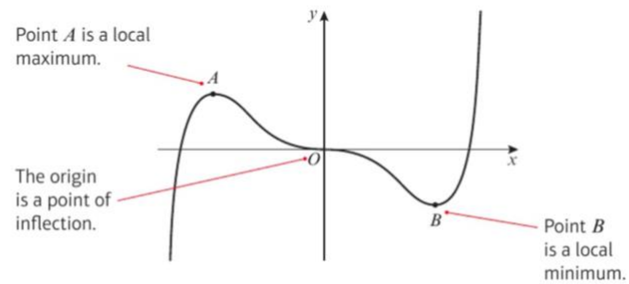
Type of stationary point	$f'(x-h)$	$f'(x)$	$f'(x+h)$
Local maximum	Positive	0	Negative
Local minimum	Negative	0	Positive
Point of inflection	Negative	0	Negative
	Positive	0	Positive



$\Rightarrow$ ***If a function $f(x)$ has stationary point when $x = a$,***

- *if $f''(a) > 0$, then it is a local minimum.*
- *if $f''(a) < 0$, then it is a local maximum.*
- *if $f''(a) = 0$, then it could be a local minimum, maximum or point of inflection.*

$y = f(x)$	$y = f'(x)$
Maximum or minimum	Cuts the x-axis
Point of inflection	Touches the x-axis
Positive gradient	Above the x-axis
Negative gradient	Below the x-axis
Vertical asymptote	Vertical asymptote
Horizontal asymptote	Horizontal asymptote at the x-axis

$\Rightarrow$ ***Definite integral*** *– calculating an integral between two limits.*

$\Rightarrow$ *Definite integrals provide a* ***value*** *whereas indefinite integral provides a* ***function***.

$\Rightarrow$ *Stationary point on a curve is any point where the curve has gradient zero.*

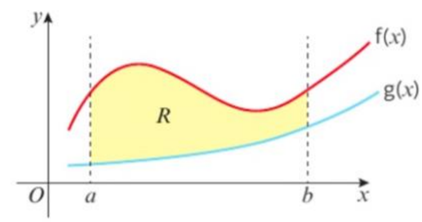
$\Rightarrow$ $\int_a^b f'(x)dx = [f(x)]_a^b = f(b) - f(a)$

$\Rightarrow$ ***The area between a positive curve, the x – axis and lines $x = a$ and $x = b$ is:***

- *Area* $= \int_a^b f(x)\,dx$

$\Rightarrow$ *When the a**rea bounded by a curve is below the x – axis,*** $\int y\,dx$ *gives a negative answer.*

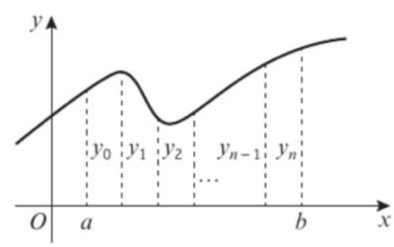
Area of $\boldsymbol{R} = \int_a^b [\mathbf{f}(x) - \mathbf{g}(x)]\,\mathbf{d}x = \int_a^b \mathbf{f}(x)\,\mathbf{d}x - \mathbf{g}(x)\,\mathbf{d}x$



$\Rightarrow$ ***Trapezium rule*** –

- $\int_a^b f(x)dx = \frac{1}{2}h(y_0 + 2(y_1 + y_2 + \cdots + y_{n-1}) + y_n)$

where $h = \dfrac{b-a}{n}$ *and* $y_i = f(a + ih)$



Statistics 1 (Formula sheet)

⇒ ***For mutually exclusive events***,
$P(A \text{ or } B) = P(A) + P(B)$

⇒ ***For independent events***,
$P(A \text{ and } B) = P(A) \times P(B)$

⇒ *The event* (***A and B***) *can be written as* $\boldsymbol{A \cap B}$.

⇒ *The event* (***A or B***) *can be written as* $\boldsymbol{A \cup B}$.

⇒ *The event* (***not A***) *can be writen as* $\boldsymbol{A'}$.

⇒ *The probability that* ***B occurs given that A has already occurred*** *is written as* $\boldsymbol{P(B|A)}$.

⇒ ***For independent events***,
- $P(A|B) = P(A|B') = P(A)$
- $P(B|A) = P(B|A') = P(B)$

⇒ ***Addition rule***:
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

⇒ ***Conditional probability***:
$$P(A \text{ given } B) = P(A|B) = \frac{P(A \cap B)}{P(B)}$$

⇒ ***Multiplication rule***:
$$P(A \cap B) = P(A|B) \, x \, P(B) \text{ or } P(B|A) \, x \, P(A)$$

⇒ ***A and B are independent if***:
- $P(A|B) = P(A)$
- $P(B|A) = P(B)$
- $P(A \cap B) = P(A) \, x \, P(B)$

⇒ ***A and B are mutually exclusive if***:
- $P(A \cap B) = 0$

⇒ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

- ***A probability distribution can be written as a***:

⇒ *Example of a fair dice rolled*:

- ***Probabilty function***:
$P(X = x) = \frac{1}{6}, x = 1,2,3,4,5,6$
- ***Table***:

X	1	2	3	4	5	6
P(X=x)	1/6	1/6	1/6	1/6	1/6	1/6

- ***Diagram***

⇒ *When all probabilities are the* ***same***, *it is known as* ***discrete unifrom distribution***.

⇒ *Sum of the probabilities of all outcomes add upto* 1.

⇒ ***Cumulative distribution*** –
$F(x) = P(X \leq x)$

- *Can be found by adding all the probabilites for those outcomes than are less than or equal to x.*

- *Example: Two fair coins are tossed. X is number of heads shown on the two coins.*

No. of heads x	0	1	2
P(X=x)	0.25	0.5	0.25
F(x)	0.25	0.75	1

⇒ ***Expected value***:
- $E(X) = \sum xP(X = x)$
- $E(X^2) = \sum x^2 P(X = x)$
- $Var(X) = E(X^2) - \left(E(X)\right)^2$
- $Var(aX + b) = a^2 Var(x)$
- $E(aX + b) = aE(X) + b$

⇒ ***For a discrete uniform distribution***:
- $E(X) = \frac{n+1}{2}$
- $Var(X) = \frac{(n+1)(n-1)}{12}$

⇒ *A* ***continuous random variable*** *has a continuous probability distribution which can be shown on a graph.*

⇒ *The* ***area*** *under a continuous probability distribution curve is equal to* **1**.

⇒ $\boldsymbol{\mu}$ *represents* ***mean***

⇒ $\boldsymbol{\sigma^2}$ *represents* ***variance***

⇒ *Symmetrical, bell – shaped curve with asymptotes at each end*

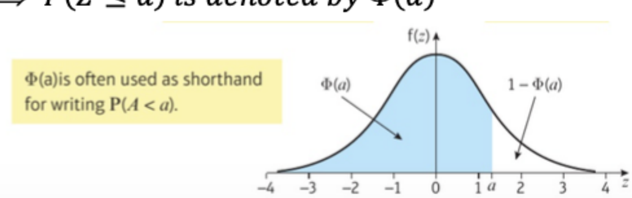
⇒ ***Standard normal variable*** *is denoted by Z, has a mean of 0 and standard deviation of 1.*
- $Z \sim N(0,1^2)$
- $\mu = 0 \; \& \; \sigma^2 = 1$

⇒ ***Normal random variable*** –
$X \sim N(\mu, \sigma^2)$

⇒ $P(Z < -a) = 1 - P(Z - a)$

⇒ $P(Z > -a) = P(Z < a)$

⇒ $P(Z \leq a)$ *is denoted by* $\Phi(a)$



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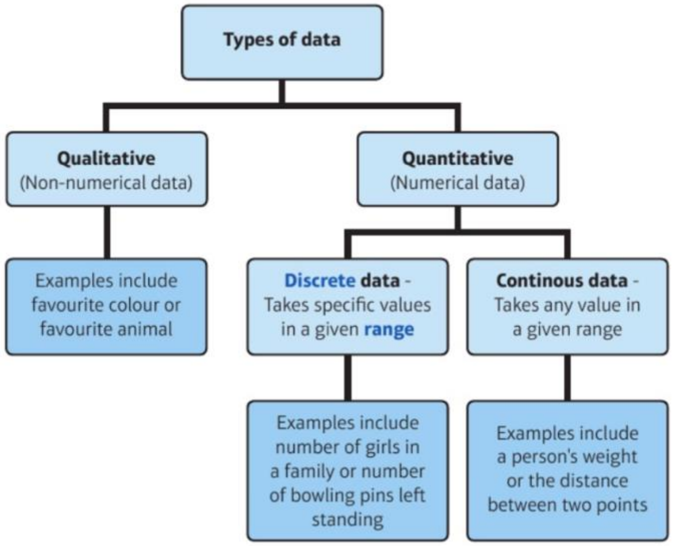
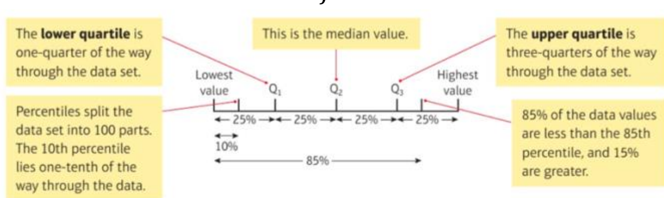


⇒ ***Large amounts of discrete*** *data can be written as a* ***frequency table*** *or as grouped data.*

⇒ *The* **number** *of anything is called its* ***frequency****, where f stands for frequency.*

⇒ ***Measure of location*** *– a single value which describes a position in a data set.*

- <u>*Measure of central tendency – the*</u> *single value describes the center of the data.*
  - *Mode – most occurred value*
  - *Median – middle value when data is put in order*
  - *Mean* $- \dfrac{\textit{Sum of the data values}}{\textit{Number of data values}}$



⇒ ***For grouped continuous data***:

- $Q_1 = \frac{n}{4}$ *th data value*
- $Q_2 = \frac{n}{2}$ *th data value*
- $Q_3 = \frac{3n}{4}$ *th data value*

⇒ *When data is presented in a grouped frequency table, we can use* ***interpolation*** *to find median, quartiles and percentiles.*

⇒ ***Measure of spread*** *– measure of how spread out the data are.*

- *Range – Difference between the largest and smallest values in the data set.*
- *Interquartile range (IQR) – difference between the upper quartile and the lower quartile.*
- *Interpercentile range – Difference between the values for the given percentiles.*

⇒ ***Variance*** $= \dfrac{\sum(x-\bar{x})^2}{n} = \dfrac{\sum x^2}{n} - \left(\dfrac{\sum x}{n}\right)^2 = \dfrac{S_{xx}}{n}$

- *where* $S_{xx} = \sum(x-\bar{x})^2 = \sum x^2 - \dfrac{(\sum x)^2}{n}$

⇒ ***Standard deviation*** $\left\{\begin{matrix}\textbf{square root of}\\ \textbf{variance}\end{matrix}\right\}$:

- $\sigma = \sqrt{\dfrac{\sum(x-\bar{x})^2}{n}} = \sqrt{\dfrac{\sum x^2}{n} - \left(\dfrac{\sum x}{n}\right)^2} = \sqrt{\dfrac{S_{xx}}{n}}$

⇒ <u>*For grouped data:*</u>

- $\sigma = \sqrt{\dfrac{\sum f(x-\bar{x})^2}{\sum f}} = \sqrt{\dfrac{\sum fx^2}{\sum f} - \left(\dfrac{\sum fx}{\sum f}\right)^2}$
- $\sigma^2 = \dfrac{\sum f(x-\bar{x})^2}{\sum f} = \dfrac{\sum fx^2}{\sum f} - \left(\dfrac{\sum fx}{\sum f}\right)^2$

⇒ ***Coding*** *– way of simplifying statistical calculations.*

- <u>*If data are coded using the formula*</u>
$$y = \frac{x-a}{b}$$
- <u>*the mean of the coded data is given by*</u>
$$\bar{y} = \frac{\bar{x}-a}{b}$$
- <u>*the standard deviation of the coded data*</u> *is given by* $\sigma_y = \dfrac{\sigma_x}{b}$*, where* σ_x *is the standard deviation of the original data.*
- ***Mean*** $= \{\bar{x} = b\bar{y} + a\}$
- ***Standard deviation*** $\{\sigma_x = b\sigma_y\}$

⇒ *Grouped continuous data can be represented in a* ***histogram****.*

⇒ *In a histogram, to calculate the height of each bar (the frequency density) use the formula:*

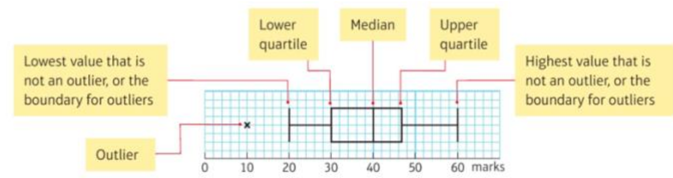
- ***area of bar*** $=$ *k x frequency*
- ***Frequency density*** $= \dfrac{\textit{Frequency}}{\textit{Class width}}$

⇒ *Joining the middle of the top of each bar in a histogram forms a* ***frequency polygon.***

⇒ ***Outlier*** *– extreme value that lies outside the overall pattern of the data.*

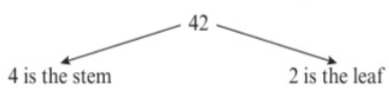
- *either greater than* $Q_3 + k(Q_3 - Q_1)$
- *or less than* $Q_1 - k(Q_3 - Q_1)$

⇒ ***Box plot*** *– shows the quartiles, any outliers, maximum and minimum values.*

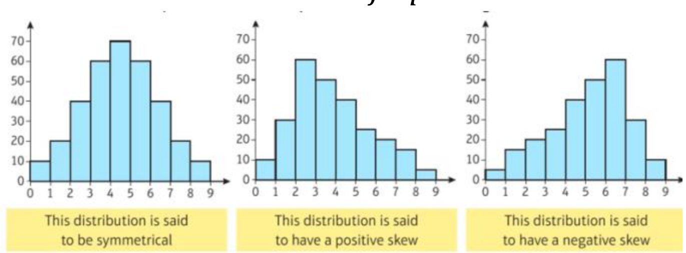


Two sets of data can be compared using box plots.

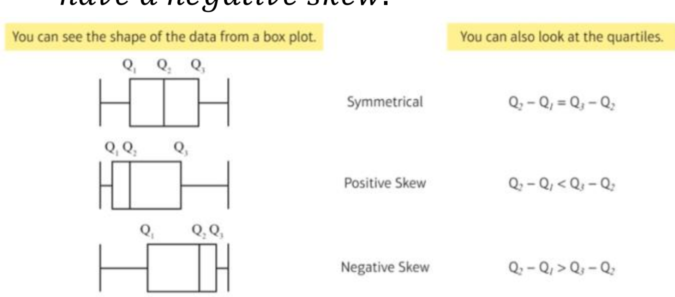
⇒ ***Stem and leaf diagram*** *– order and present data given to 2 or 3 significant figures.*



⇒ ***Skewness*** *– The shape of a data set can be described using diagrams, measures of location and measures of spread.*



- *Data which are spread evenly are symmetrical.*
- *Data which are mostly at lower values have a positive skew.*
- *Data which are mostly at higher values have a negative skew.*



Another test uses the measures of location:

- Mode = median = mean describes a distribution which is **symmetrical**
- Mode < median < mean describes a distribution with a **positive skew**
- Mode > median > mean describes a distribution with a **negative skew**

You can also calculate $\frac{3(\text{mean} - \text{median})}{\text{standard deviation}}$ which tells you how **skewed** the data are.

- A value of 0 implies that the mean = median and the distribution is **symmetrical**
- A positive value implies that the median < mean and the distribution is **positively skewed**
- A negative value implies that median > mean and the distribution is **negatively skewed**

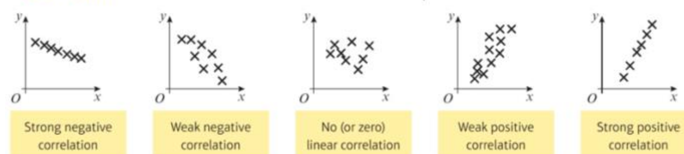
The further from 0 the value is, the more likely the data will be skewed.

⇒ ***When comparing data sets****, comment on:*

- *Measures of location*
- *Measures of spread*

⇒ ***Bivariate data*** *are data which have pairs of values for two variables. It can be represented on a scatter diagram.*

- **Correlation** describes the nature of the linear relationship between two variables.



⇒ *When a scatter diagram shows correlation, you can draw* ***least squares regression line*** *[straight linethat minimises the sum of the squares of the vertical distances of each data point from the line].*

- ***Regression line***: $y = a + bx$

where $b = \frac{S_{xy}}{S_{xx}}$ *and* $a = \overline{y} = b\overline{x}$

- $S_{xy} = \sum xy - \frac{\sum x \sum y}{n}$
- $S_{xx} = \sum x^2 - \frac{(\sum x)^2}{n}$
- $S_{yy} = \sum y^2 - \frac{(\sum y)^2}{n}$

⇒ ***Product moment correlation coefficient (PMCC)*** *– measures the linear correlation between two variables.*

$$r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$$

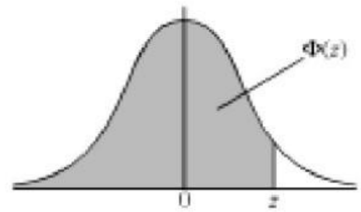
- $r = 1$ *{perfect positive linear correlation}*
- $r = -1$ *{perfect negative linear*
- *correlation}*
- $r = 0$ *{no linear correlation}*

THE NORMAL DISTRIBUTION FUNCTION

If Z has a normal distribution with mean 0 and variance 1, then, for each value of z, the table gives the value of $\Phi(z)$, where

$$\Phi(z) = P(Z \leqslant z).$$

For negative values of z, use $\Phi(-z) = 1 - \Phi(z)$.



z	0	1	2	3	4	5	6	7	8	9	1	2	3	4	5	6	7	8	9
															ADD				
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359	4	8	12	16	20	24	28	32	36
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753	4	8	12	16	20	24	28	32	36
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141	4	8	12	15	19	23	27	31	35
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517	4	7	11	15	19	22	26	30	34
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879	4	7	11	14	18	22	25	29	32
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224	3	7	10	14	17	20	24	27	31
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549	3	7	10	13	16	19	23	26	29
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852	3	6	9	12	15	18	21	24	27
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133	3	5	8	11	14	16	19	22	25
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389	3	5	8	10	13	15	18	20	23
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621	2	5	7	9	12	14	16	19	21
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830	2	4	6	8	10	12	14	16	18
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015	2	4	6	7	9	11	13	15	17
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177	2	3	5	6	8	10	11	13	14
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319	1	3	4	6	7	8	10	11	13
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441	1	2	4	5	6	7	8	10	11
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545	1	2	3	4	5	6	7	8	9
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633	1	2	3	4	4	5	6	7	8
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706	1	1	2	3	4	4	5	6	6
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767	1	1	2	2	3	4	4	5	5
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817	0	1	1	2	2	3	3	4	4
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857	0	1	1	2	2	2	3	3	4
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890	0	1	1	1	2	2	2	3	3
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916	0	1	1	1	1	2	2	2	2
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936	0	0	1	1	1	1	1	2	2
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952	0	0	0	1	1	1	1	1	1
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964	0	0	0	0	1	1	1	1	1
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974	0	0	0	0	0	1	1	1	1
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981	0	0	0	0	0	0	0	1	1
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986	0	0	0	0	0	0	0	0	0

Percentage Points Of The Normal Distribution

The values z in the table are those which a random variable $Z \sim N(0, 1)$ exceeds with probability p; that is, $P(Z > z) = 1 - \Phi(z) = p$.

p	z	p	z
0.5000	0.0000	0.0500	1.6449
0.4000	0.2533	0.0250	1.9600
0.3000	0.5244	0.0100	2.3263
0.2000	0.8416	0.0050	2.5758
0.1500	1.0364	0.0010	3.0902
0.1000	1.2816	0.0005	3.2905

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